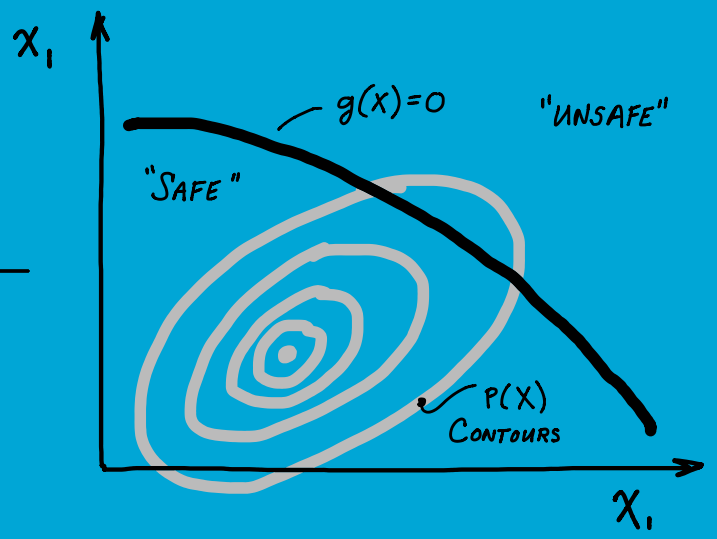
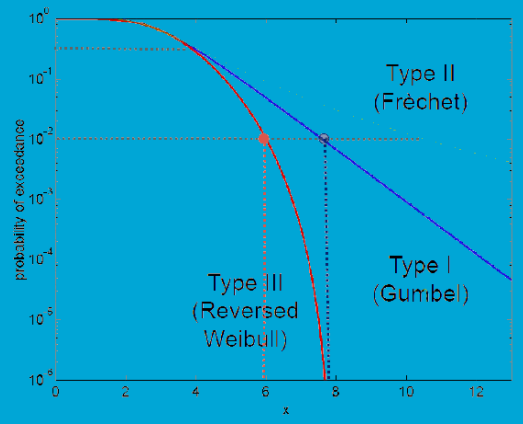
CIEM42X0 Probabilistic Design

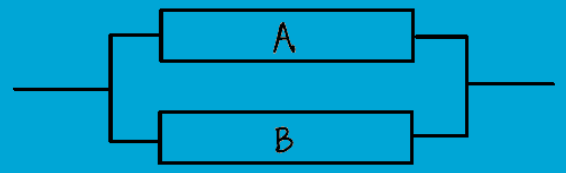
Hydraulic and Offshore Structures (HOS) Track

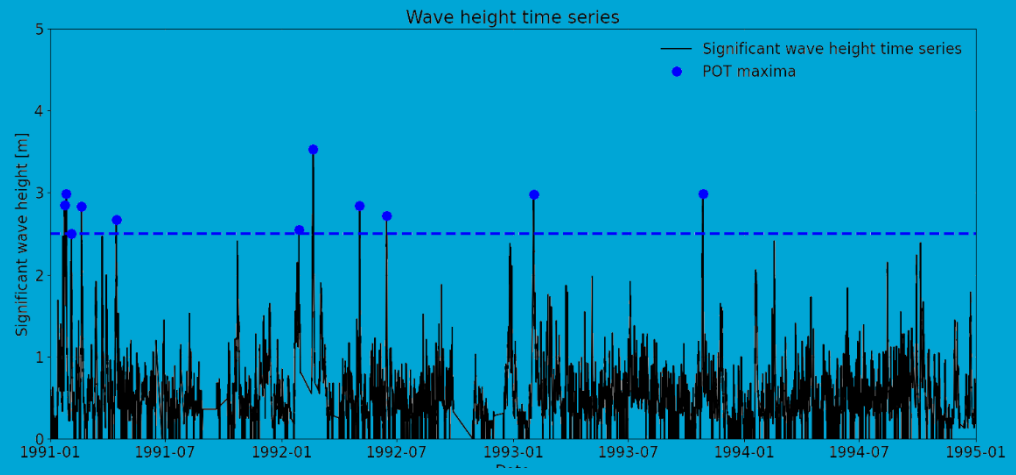
Civil Engineering MSc Program



Introduction - Logistics







Patricia Mares Nasarre

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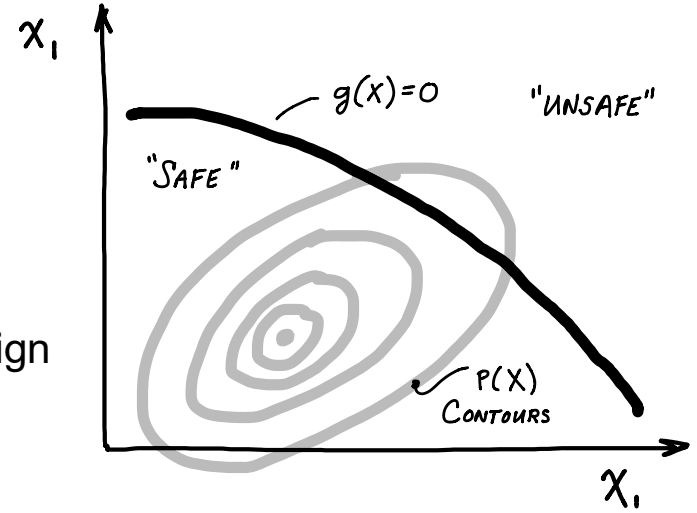
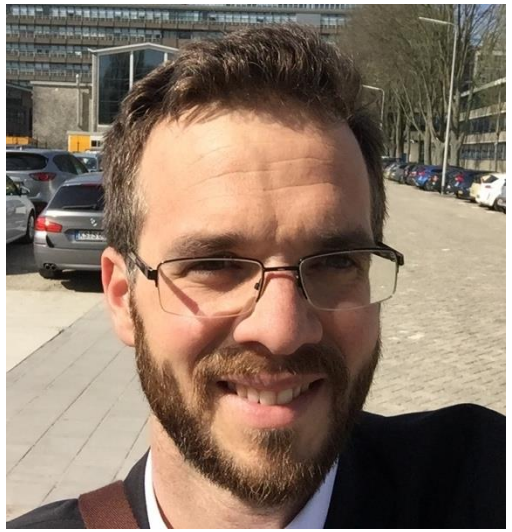


3 main topics: extreme value analysis, component reliability, system reliability

Objective: learn how the methods can be used to gain insight and improve your design

Patricia Mares Nasarre
Extreme value analysis,
Component Reliability

Oswaldo Morales Nápoles
System Reliability



Robert Lanzafame
Honorific mention

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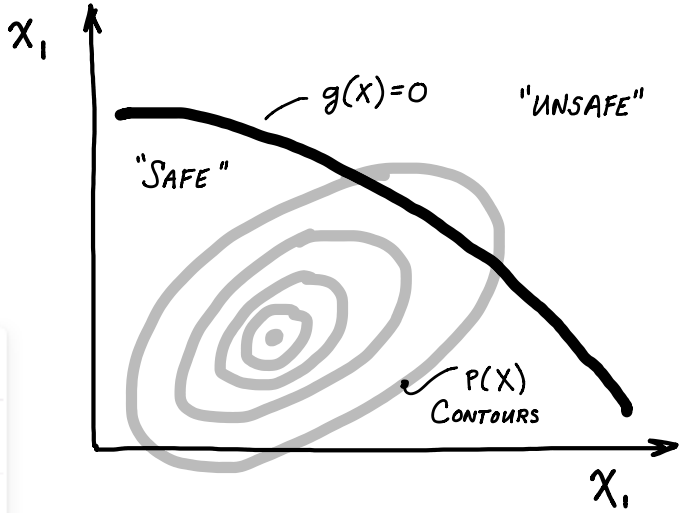

Course setup: holidays make it a mess!

Week	Date	Time	Content
1	20/4	15.45 - 17.30h	Introduction & recap
1	24/4	9.45 - 12.30h	Exercise: recap reliability
2	28/4	15.45 - 17.30h	Theory: Extension on Extreme Value Analysis
2	1/5	9.45 - 12.30h	Exercise*: Extension on Extreme Value Analysis
3	4/5	15.45 - 17.30h	Theory: Component reliability
3	8/5	9.45 - 12.30h	Exercise*: Component reliability
4	11/5	10.45 - 11.30h	Theory: System reliability
5	21/5	13.45 - 15.30h	Exercise: System reliability
5	22/5	15.45 - 17.30h	Theory: Discrete Bayesian Networks
6	29/5	9.45 - 12.30h	Exercise*: Discrete Bayesian Networks
7	5/6	9.45 - 10.30h	Q&A
8	9/6	15.45 - 16.30h	Q&A

Exercises with *: graded for students from Dams, Dikes and Breakwaters.

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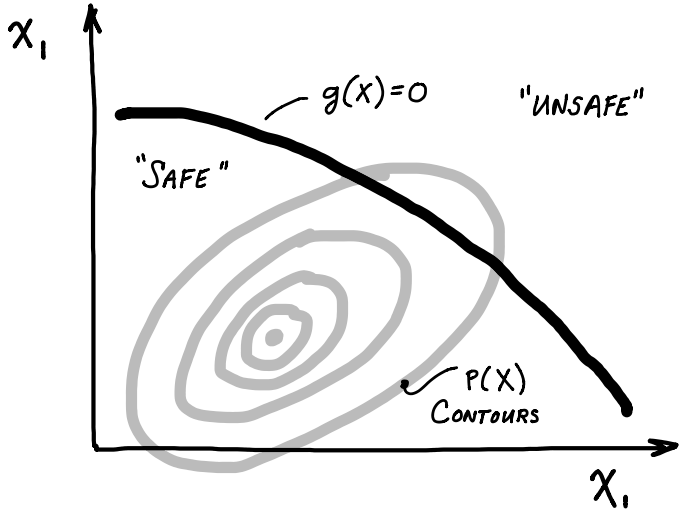

Unit Website(s): textbook links, calendar, announcements, logistics, etc

https://tudelft-citg.github.io/HOS-prob-design-26/

Assessment: incorporated in projects and exams for each unit

→ use code from assignments!

You can do <u>ALL</u> the assignments in a team (<u>THE SAME</u>) or individually.

Contact Patricia Mares Nasarre for questions.

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Demonstration (not on slides)

- Website
- Online Textbook

- This week
  - Review MUDE
  - Set up your environment

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Programming

This will be homework → Python setup ☺

- You need to set up a new environment by Friday week 2 (for week 1, MUDE env should work)
  - pyExtremes
  - OpenTURNs
  - Git**Hub** ←

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Summary

- Review MUDE weeks 1.4, 1.5, 2.7 & 2.8
- Set up your new environment
- Let Patricia know if you have questions!

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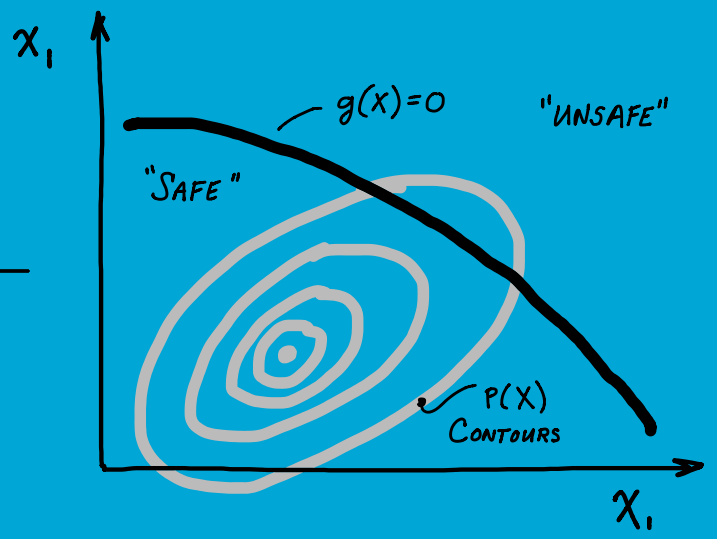
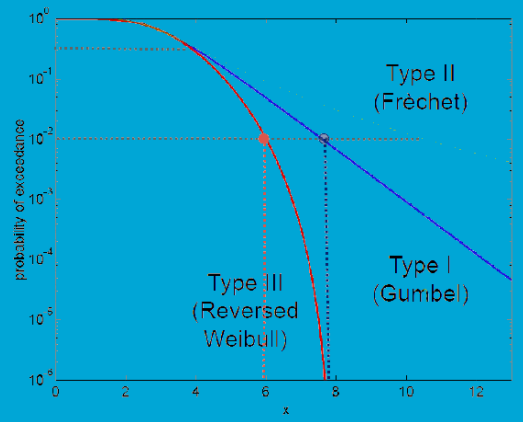
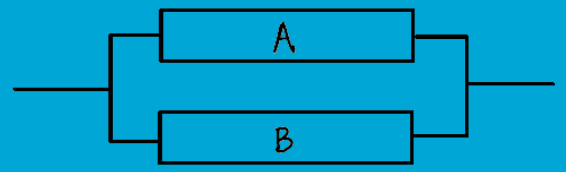
Hydraulic and Offshore Structures (HOS) Track

Civil Engineering MSc Program

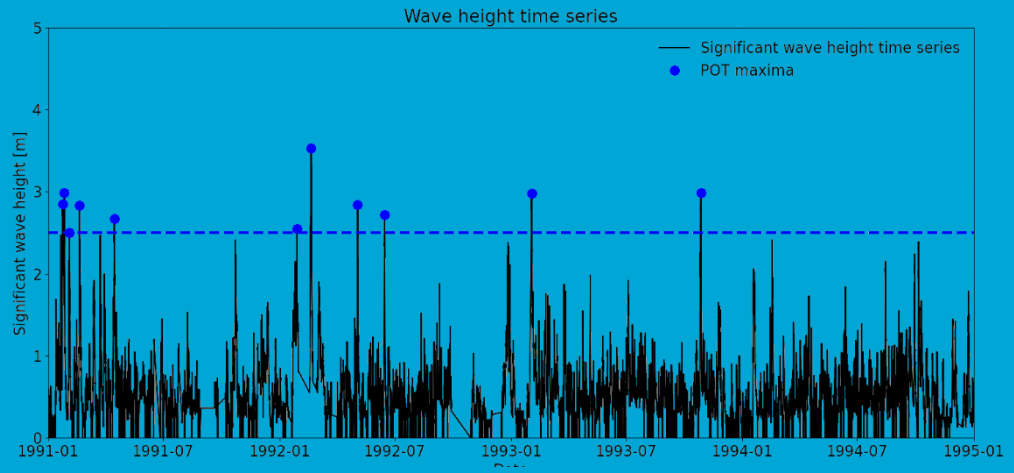
Recap on probability and reliability







Patricia Mares Nasarre



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Outline

1. **Refresher: continuous distribution functions**
   a. **Univariate**
   b. Multivariate

2. (Risk and) Reliability analysis

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1a. Univariate (continuous) distributions – What's that?

- Univariate → 1 variable
- Continuous → Continuous variables (not discrete)
- Univariate (continuous) distributions:
  - *Mathematical model which relates the values of a random variable and their probability*

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1a. Univariate (continuous) distributions - functions

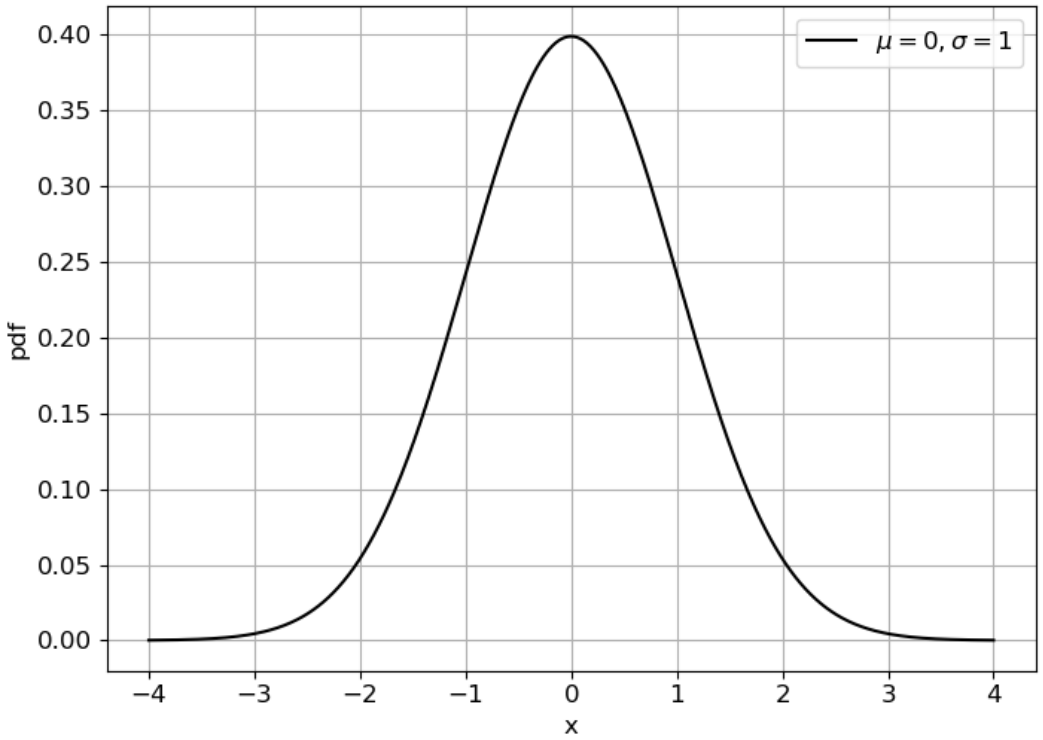
- Do you remember any functions?

- Probability **density** function (PDF) $f_X(x)$

$$f_X(x)dx = P(x < X \leq x + dx)$$

$$f_X(x) \geq 0$$

$$\int_{-\infty}^{+\infty} f_X(x)dx = 1$$



PDF of the Gaussian distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

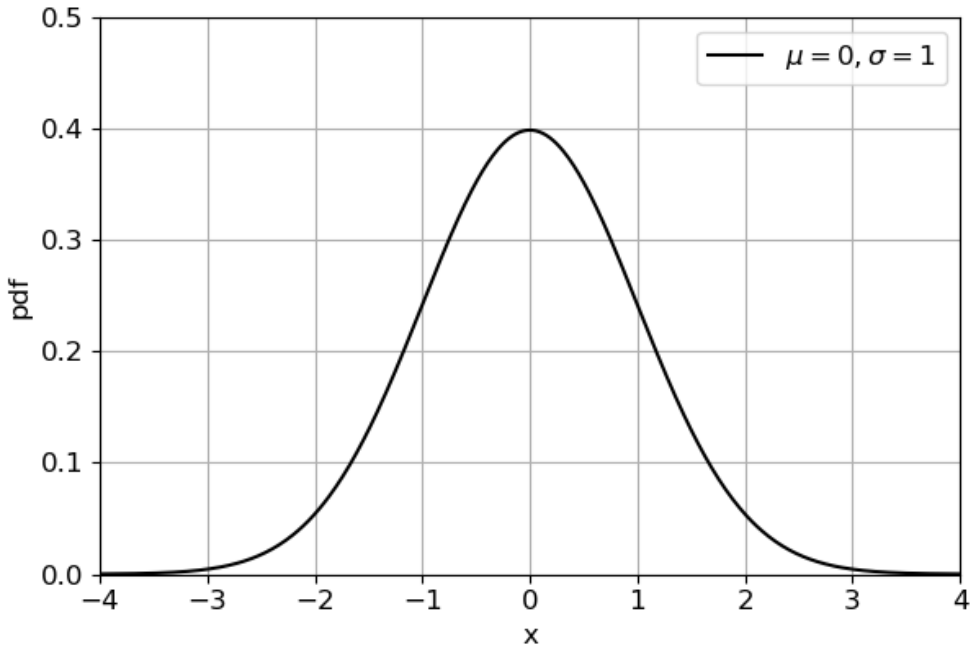
1a. Univariate (continuous) distributions - From PDF to CDF

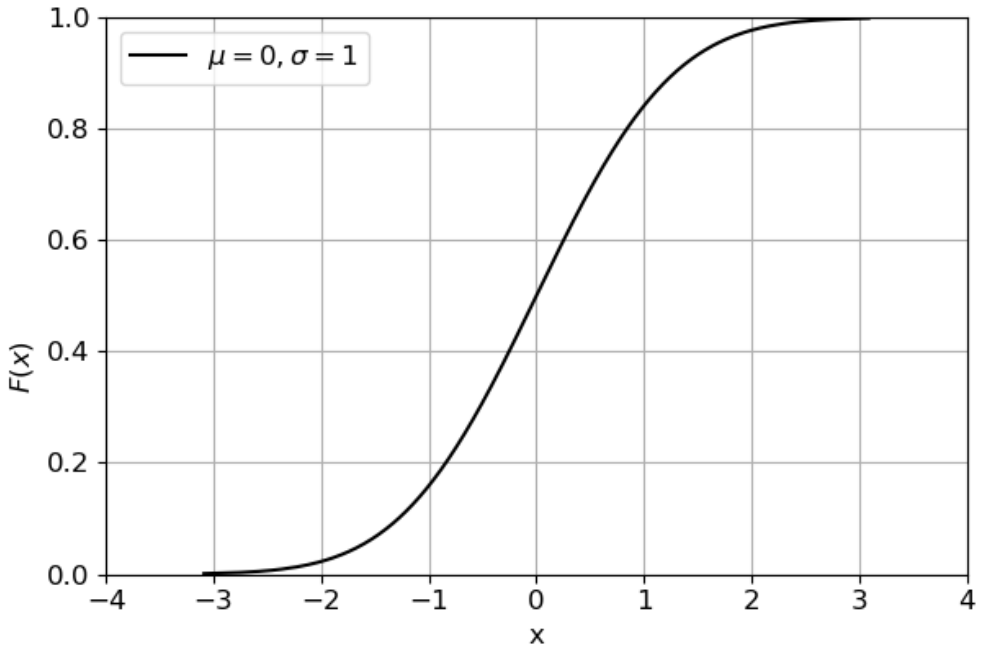
- Probability density function (PDF) $f_X(x)$

- Cumulative distribution function (CDF)

CDF of the Gaussian distribution

$$F(x) = \frac{1}{2}\left(1 + \mathrm{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)\right)$$





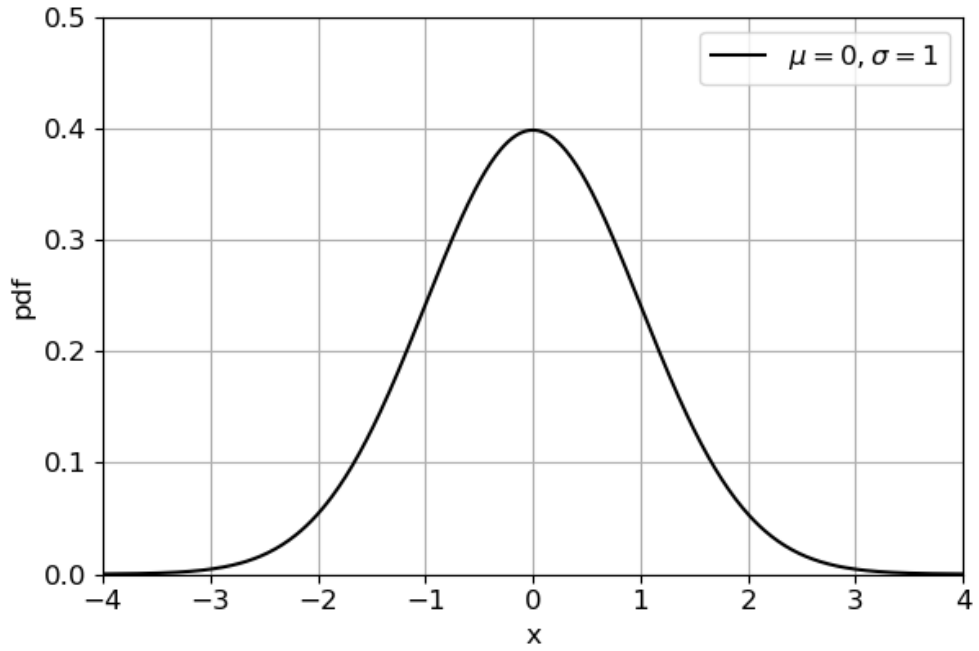
1a. Univariate (continuous) distributions - From PDF to CDF

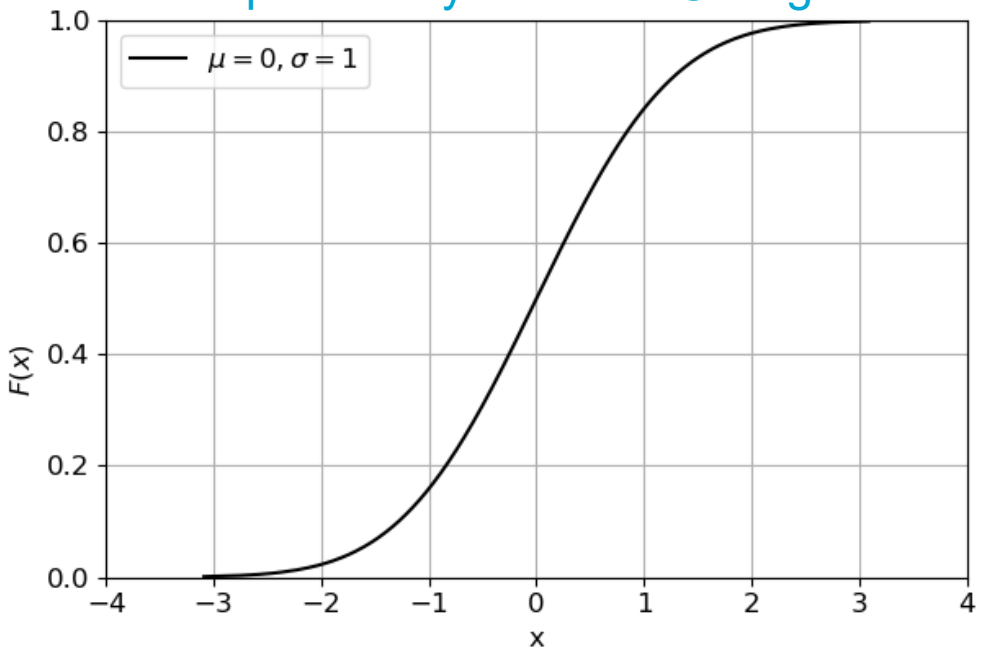
- Probability density function (PDF) $f_X(x)$

How? $F(x) = \int_{-\infty}^{x} f(x)dx$

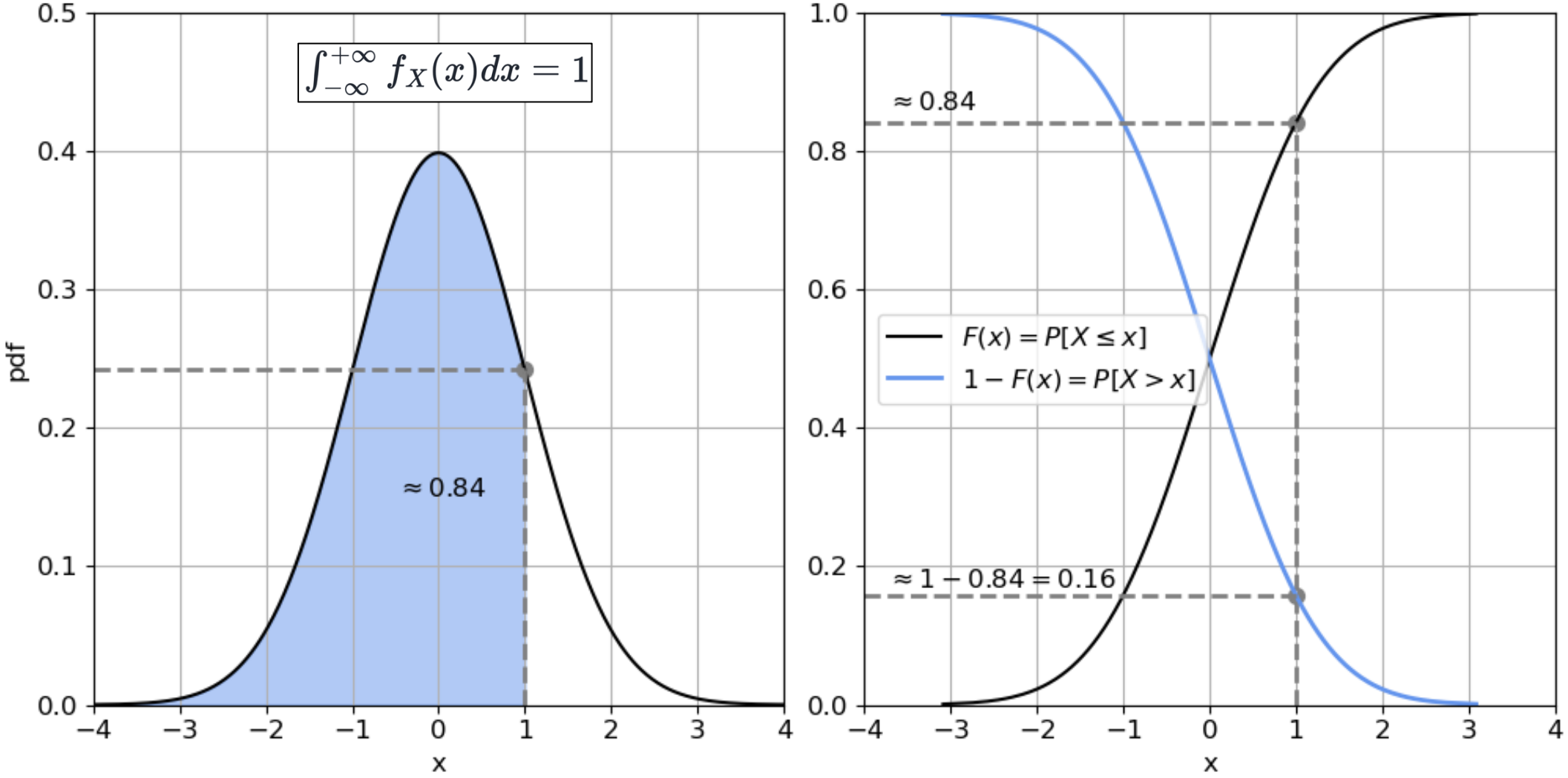
- Cumulative distribution function (CDF)

Which probability does the CDF give me?





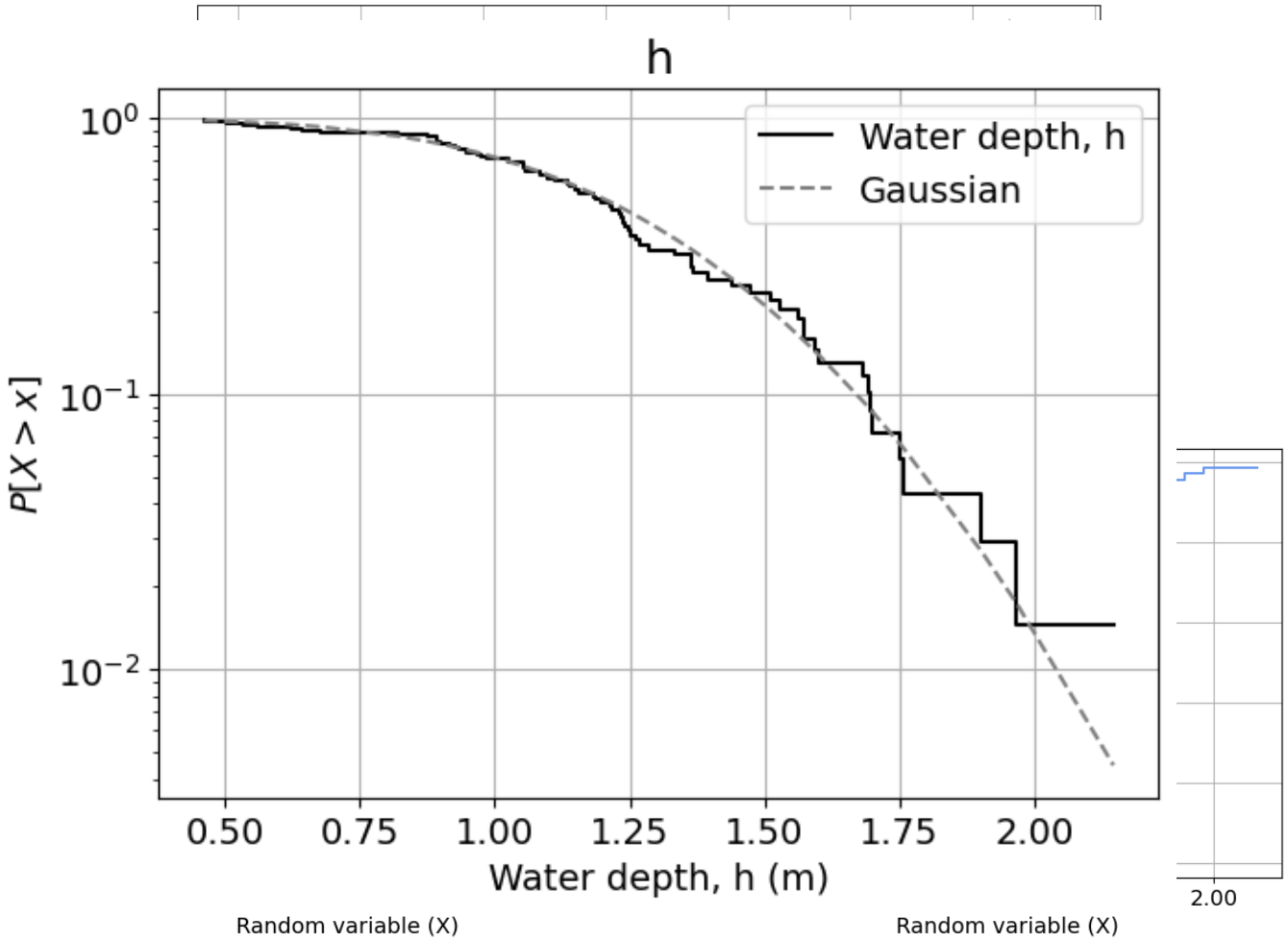
1a. Univariate (continuous) distributions– exceedance



1a. Univariate distributions– how do we fit it to data?

Given a parametric distribution function (e.g.: Gaussian distribution):

1. Build ECDF from the data
2. Fit the distribution to the ECDF using method of moments or MLE
3. Goodness of fit: graphical methods or hypothesis test
4. Is it good enough for your purposes? Then, use it!



Outline

1. **Refresher: continuous distribution functions**
   a. Univariate
   b. **Multivariate**

2. (Risk and) Reliability analysis

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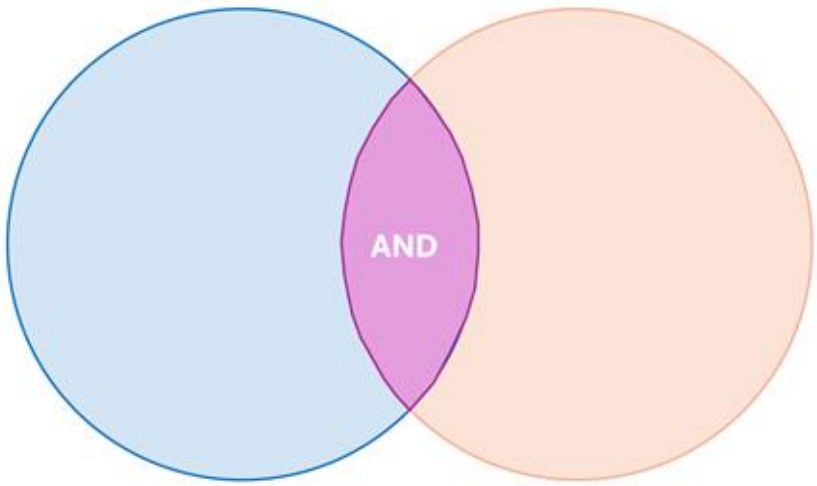
1b. Some definitions - discrete

- **Independence**:

  A and B are considered independent **if and only if** the AND probability, $P(A \cap B)$, can be factorized into the product of their probabilities.

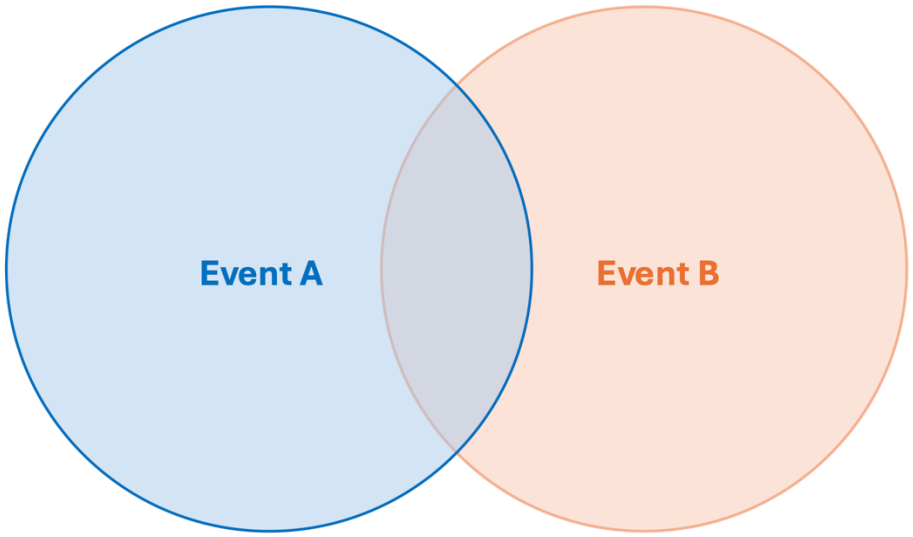
$$P(A \cap B) = P(A)P(B)$$

- $\boldsymbol{P(A \cap B)}$:



1b. Some definitions - discrete

- $\boldsymbol{P(A \cup B)}$: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$



- **Conditional probability**:

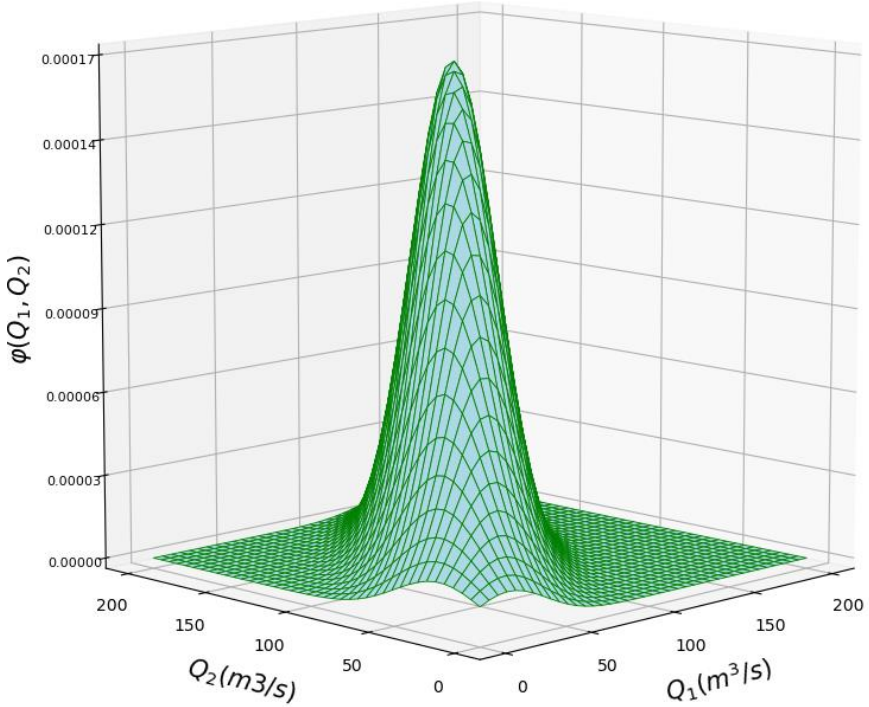
$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)\,\cancel{P(B)}}{\cancel{P(B)}} = P(A)$$

If independent

1b. Multivariate

- Univariate → 1 variable
- Continuous
- Multivariate (continuous) distributions:
  - *Mathematical model which relates the values of several random variable and their probability*

- Multivariate → > 1 variable



1b. Why multivariate?

Variables of interest are often 'tied' to each other, as they are generated by the same drivers.

An example:

- Wave height;
- Wave period;
- Wave direction;
- Wind velocity;
- Wind direction;
- Storm surge;
- and many more!

Examples:

Higher wave heights are usually associated with larger wave periods.

Wave storms usually come from the same direction(s) conditioned by the bathymetry and geography of the area. Therefore, there is a relationship between large wave heights and wave direction.

We cannot assume independence

1b. What shouldn't I do?

Toy example. I can have flooding if:

- I have high wave heights → P(Hs>3m)=0.1

AND

- I have a large storm surge (water level) → P(SS>2m)=0.07

What is my probability of flooding?

P(Hs>3m AND SS>2m) ≠ P(Hs>3m) P(SS>2m) **UNSAFE!**

1b. We need to define a multivariate distribution

- Example for **two random variables**, the *bivariate distribution* of the random variables X and Y has a PDF $f_{XY}(x,y)$ and a CDF $F_{XY}(x,y)$. The CDF is then defined as

$$\boxed{P(X \leq x, Y \leq y) = F_{XY}(x,y)} = \int_{-\infty}^{y} \int_{-\infty}^{x} f_{XY}(x,y)\,dx\,dy$$

AND!

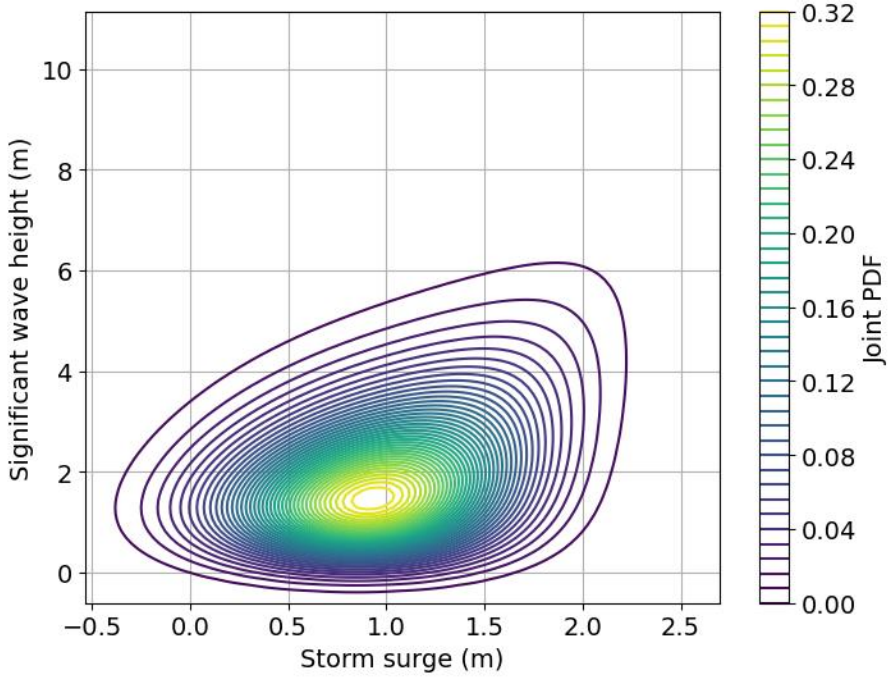
Joint probability/joint non-exceedance probability

- They can also be called *joint distributions*.

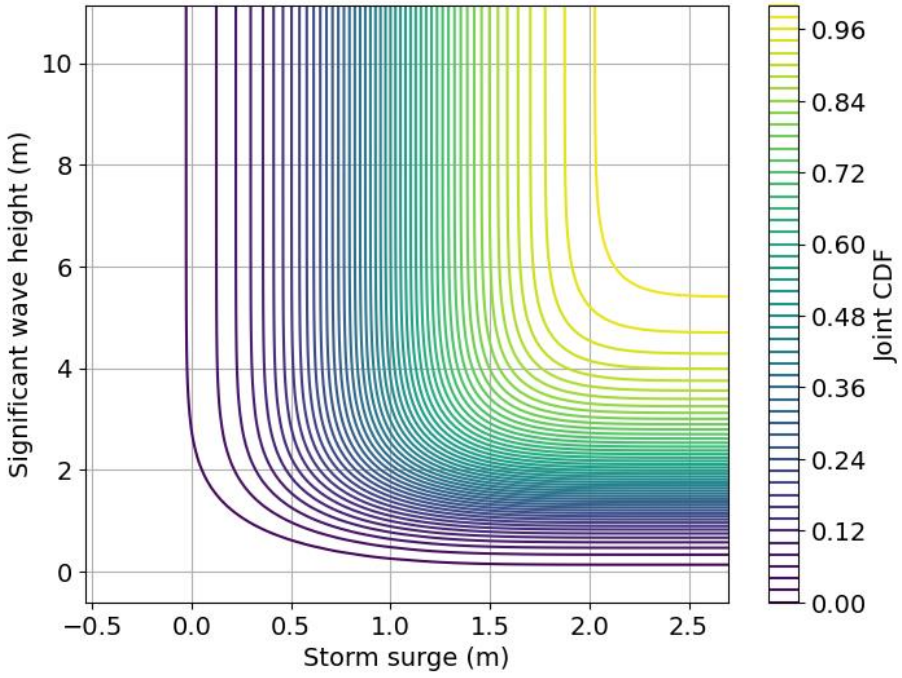
- We call *marginal distribution* to the univariate distribution associated with a single random variable that is part of a multivariate distribution.

1b. We need to define a multivariate distribution

$f_{SS,H}(ss,h)$



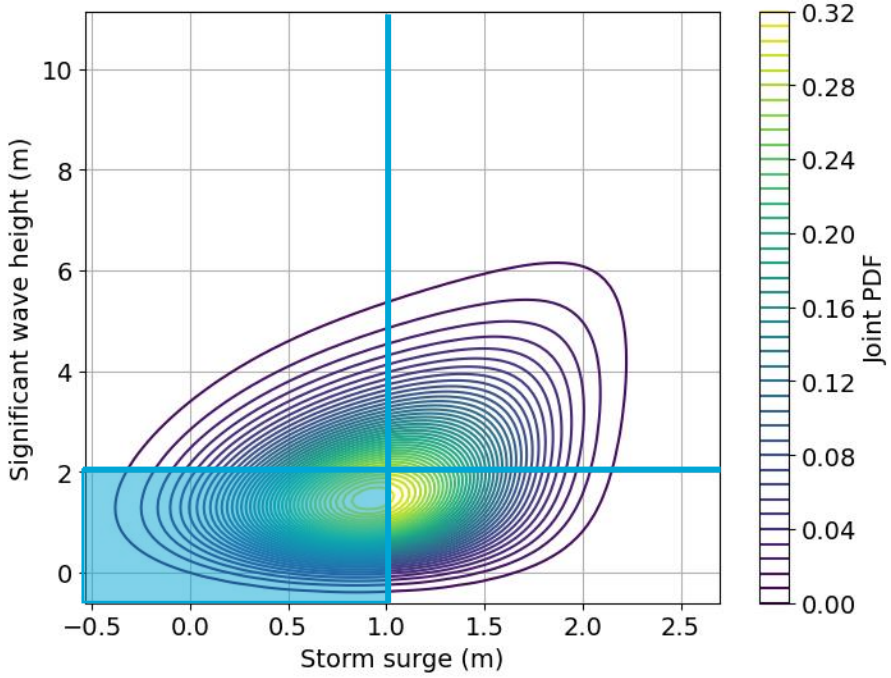
$F_{SS,H}(ss,h) = P(SS \leq ss, H \leq h)$



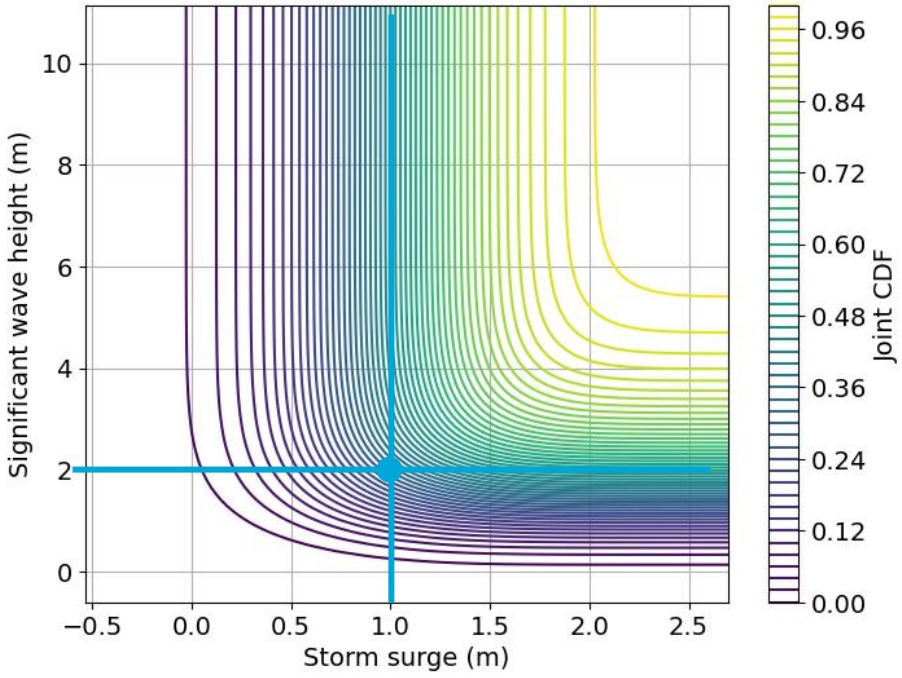
1b. We need to define a multivariate distribution

$F_{SS,H}(1,2)$ → in which region am I integrating?

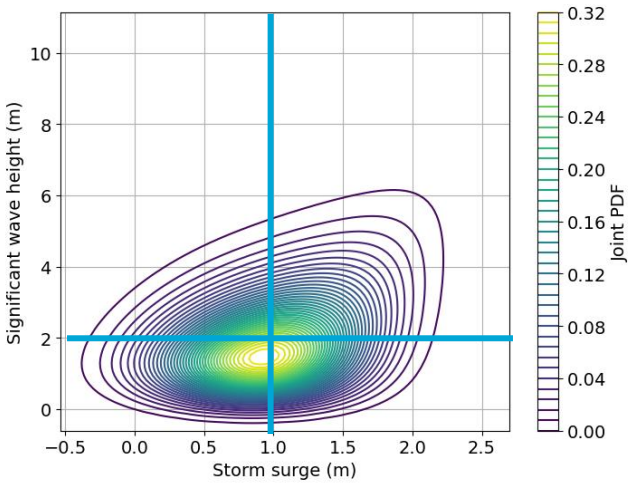
$f_{SS,H}(ss,h)$

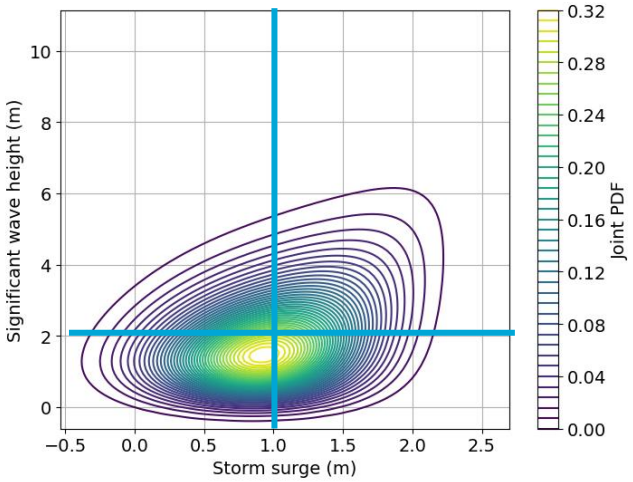


$F_{SS,H}(ss,h) = P(SS \leq ss, H \leq h)$



1b. Computations with a multivariate distribution





I know:

$F_{SS,H}(ss, h)$

$F_{SS}(ss)$

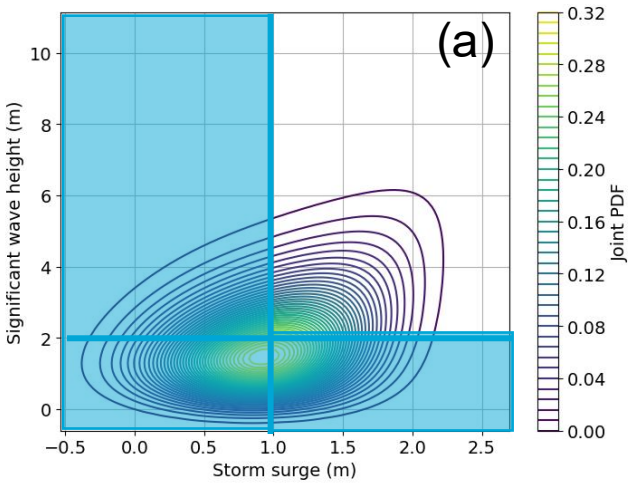
$F_H(h)$

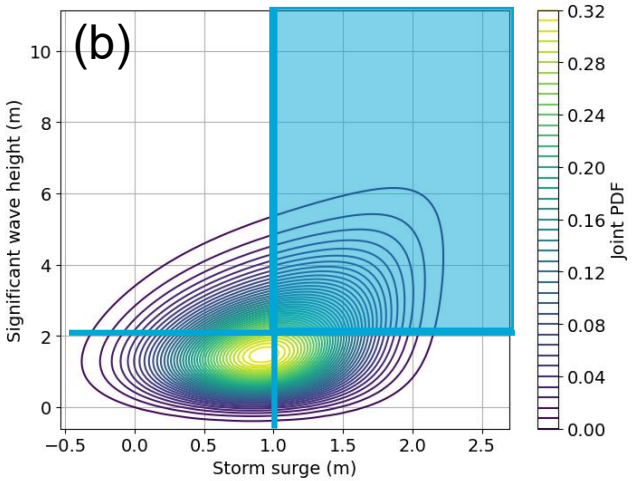
a. $P(SS \leq 1 \text{ OR } H \leq 2) =$

b. $P(SS > 1 \text{ AND } H > 2) =$

c. $P(SS > 1 \mid H > 2) =$

1b. Computations with a multivariate distribution





I know:

$F_{SS,H}(ss, h)$

$F_{SS}(ss)$

$F_H(h)$

a. $P(SS \leq 1 \text{ OR } H \leq 2) =$

$= F_{SS}(1) + F_H(2) - F_{SS,H}(1,2)$

b. $P(SS > 1 \text{ AND } H > 2) =$

$= 1 - F_{SS}(1) - F_H(2) + F_{SS,H}(1,2)$

$= 1 - P(SS \leq 1 \text{ OR } H \leq 2)$

c. $P(SS > 1 \mid H > 2) =$

$= P(SS > 1 \text{ AND } H > 2) / P(H > 2) =$

$= (1 - F_{SS}(1) - F_H(2) + F_{SS,H}(1,2)) / (1 - F_H(2))$

1b. Which multivariate distributions do you know?

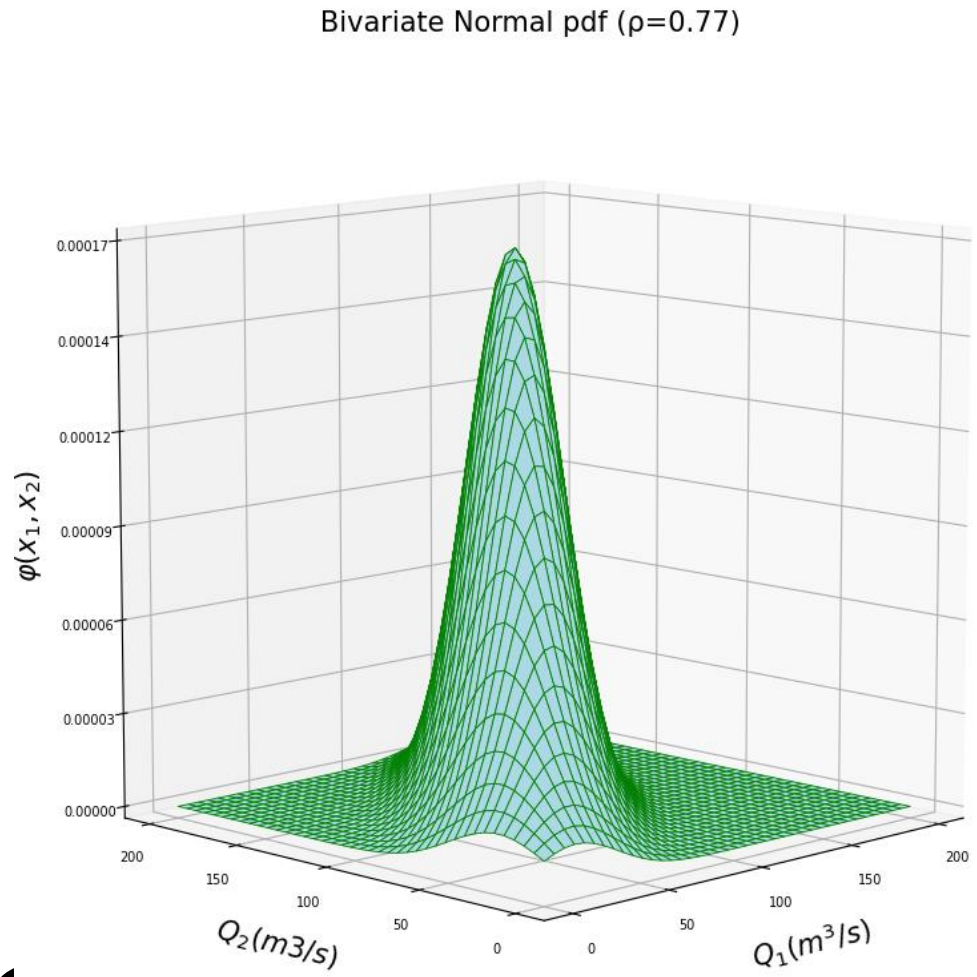
Bivariate Gaussian distribution PDF:

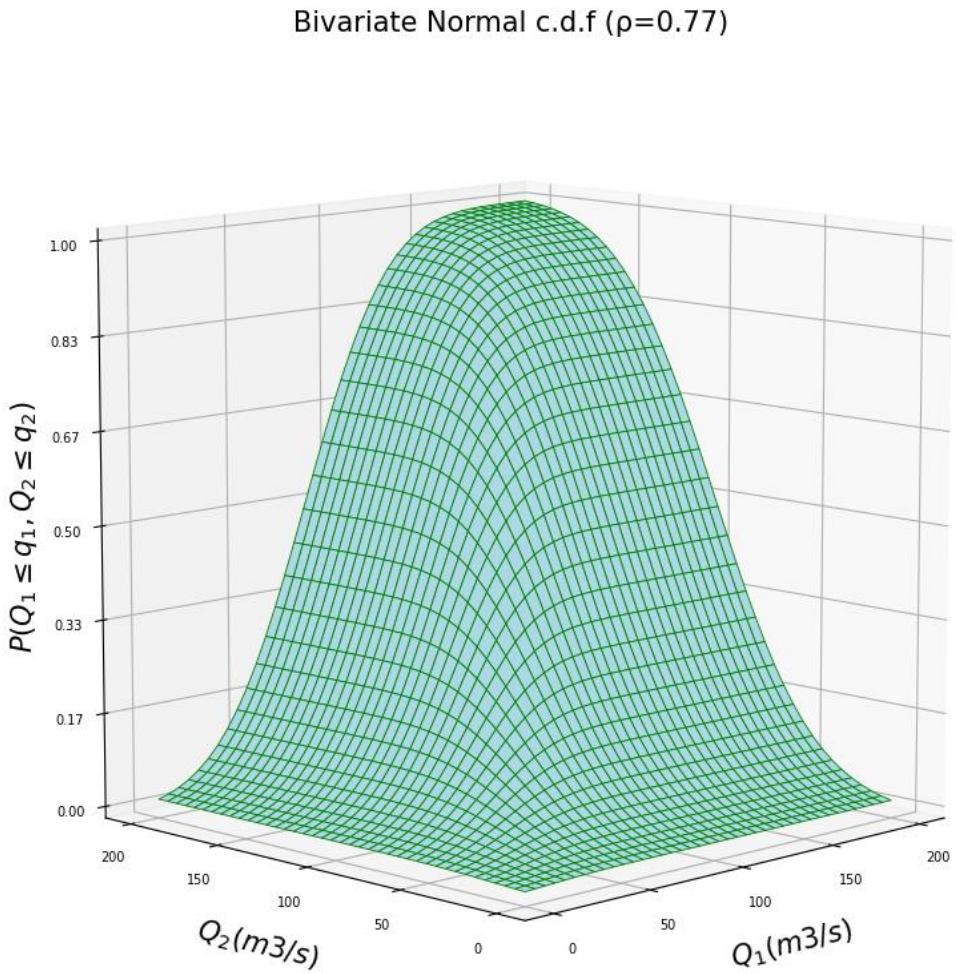
$$\phi_\rho(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left(-\frac{\left(\frac{x_1-\mu_1}{\sigma_1}\right) - \left(\frac{2\rho(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1\sigma_2}\right) + \left(\frac{x_2-\mu_2}{\sigma_2}\right)}{2(1-\rho^2)} \right)$$

I can be rewritten in matricial form as

$$\phi_\rho(x_1, x_2) = \frac{1}{\sqrt{(2\pi)^2 \begin{vmatrix} \sigma_1^2 & Cov(X_1, X_2) \\ Cov(X_1, X_2) & \sigma_2^2 \end{vmatrix}}} \exp\left(-\frac{1}{2}(x_1 - \mu_1 \; x_2 - \mu_2) \begin{pmatrix} \sigma_1^2 & Cov(X_1, X_2) \\ Cov(X_1, X_2) & \sigma_2^2 \end{pmatrix}^{-1} \begin{pmatrix} x_1 - \mu_1 \\ x_2 - \mu_2 \end{pmatrix} \right)$$

1b. Which multivariate distributions do you know?





1b. Which multivariate distributions do you know?

Multivariate Gaussian distribution:

- Symmetric
- Not a “flexible” model

Do you want to know how to build flexible multivariate distributions?

Go to **MORE**!

Cross Over on Probabilistic modelling

In this course, we will proceed with “simplifications”: either independence or MGd

Outline

1. Refresher: continuous distribution functions
   a. Univariate
   b. Multivariate

2. **(Risk and) Reliability analysis**

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2. Risk Analysis – what is risk?

"a situation involving exposure to danger" → focused on consequences

"the possibility that something unpleasant or unwelcome will happen" → focused on the probability of happening

Which definition do we typically use?

Risk is the probability of an undesired event multiplied by the consequences.

$$E(d_i) = p_i \cdot d_i$$

Expected damage, typically expressed in €/year

Expected damage for a given set of scenarios i=1,…, n

$$E(d) = \sum_{S_{i=1}}^{n} p_i \cdot d_i$$

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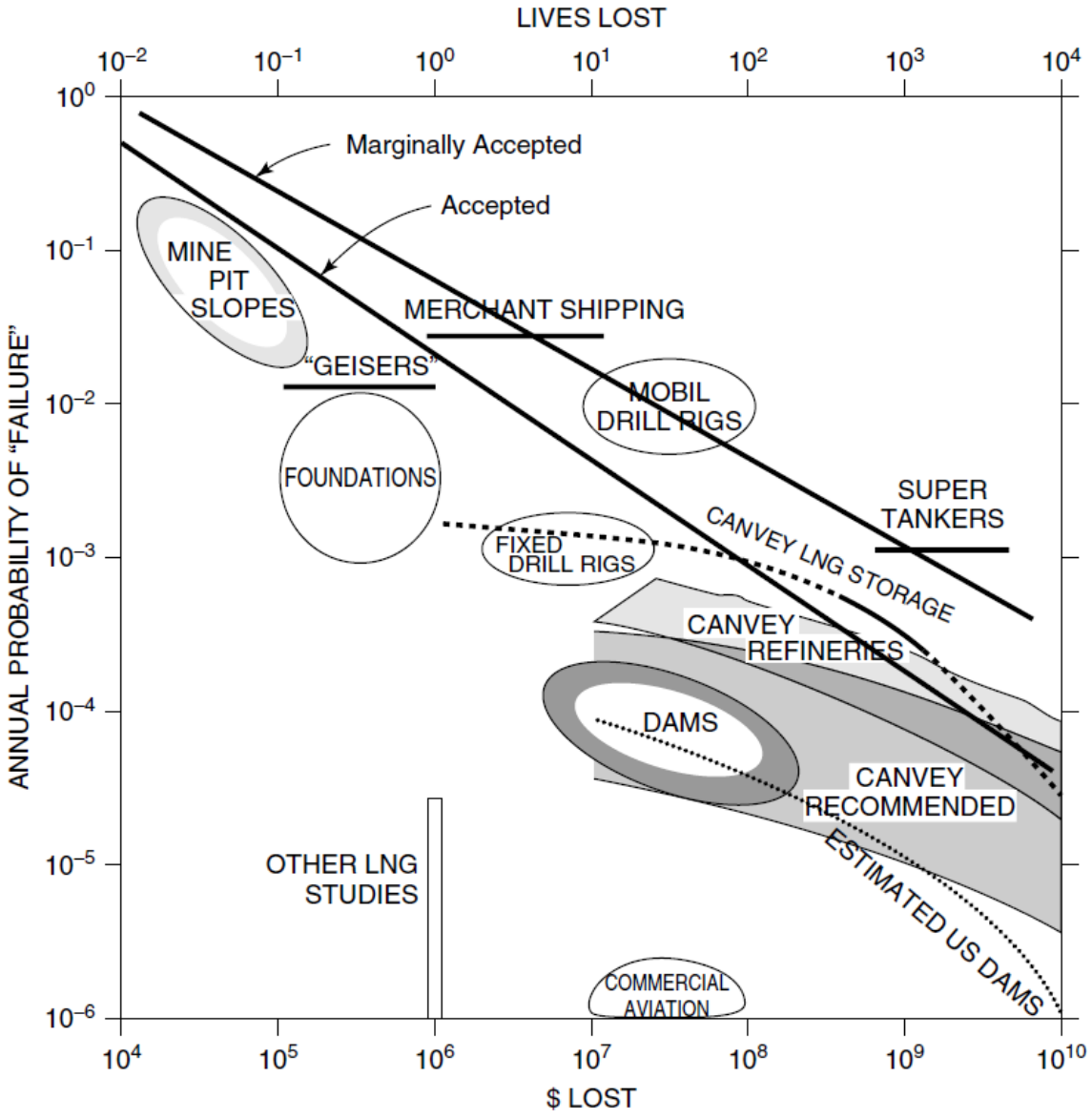
2. Risk Analysis – why?

Why do we use the concept of risk?

- Decide on acceptability
- Safety levels in the design process
- Need for measures in existing systems

Quantitative measure to transfer from decisions to technical field.

We need a probability of failure!



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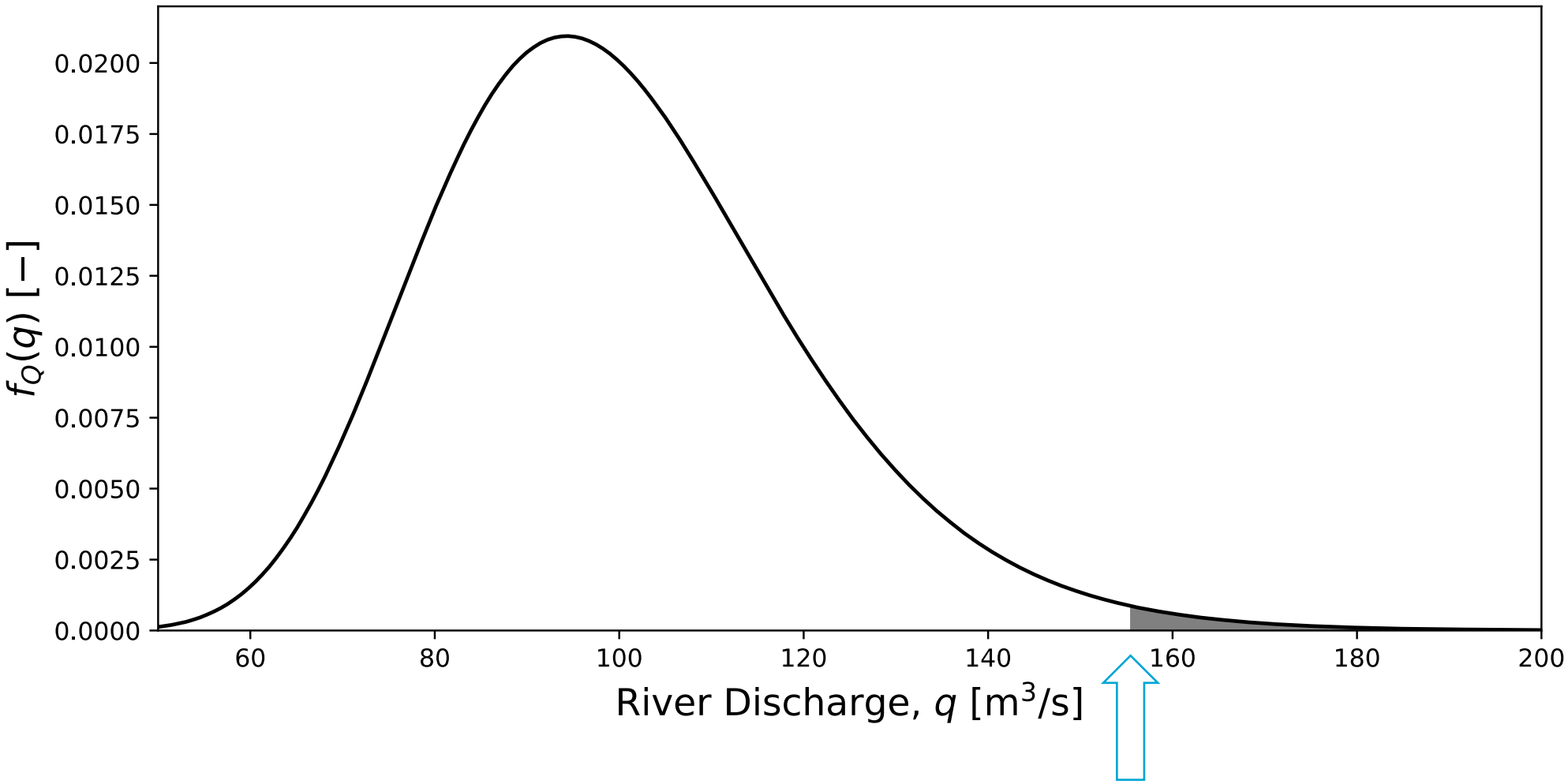
2. Reliability Analysis – how do we define failure?



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2. Reliability Analysis – how do we define failure?

We describe **failure** mathematically through a ***limit state function***, *g(x)*.

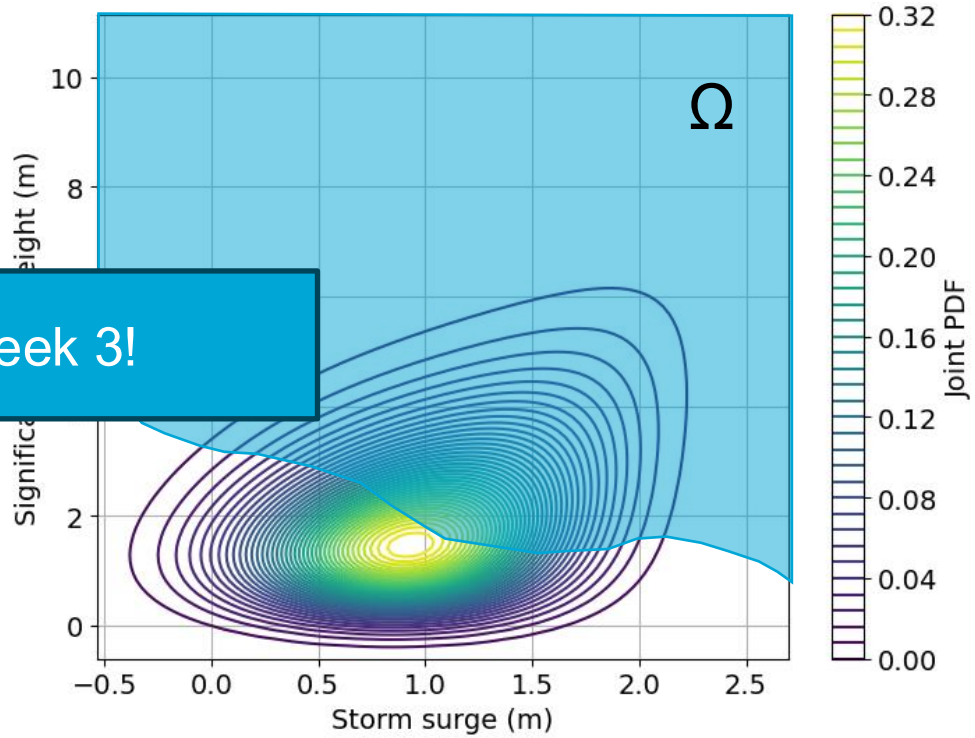
Typically, in the form of “resistance - loading”, where the probability of failure is defined as

$$P_{failure} = \mathrm{P}[g(x) < 0]$$

Toy example:

- A structure fails if received impact is larger than resistance, R;
- The magnitude of the impact is function of SS and H;
- We can define a failure region Ω;
- $P_{failure}$ is the integral over that failure region.

More in week 3!



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How we will approach reliability-based design

- Risk Analysis: evaluation, assessment, context
- Continuous Distributions, Extreme Value Analysis (partially in MUDE)
- Dependence touched lightly: discussion on implications
  (covered in cross-over!!! CEGM2005 tudelft-citg.github.io/MORE)
- Component Reliability:
  a function of random variables $p_{\mathrm{f}} = \int_{\Omega} f_X(x)\,\mathrm{d}x$
- System Reliability:
  solving complex systems

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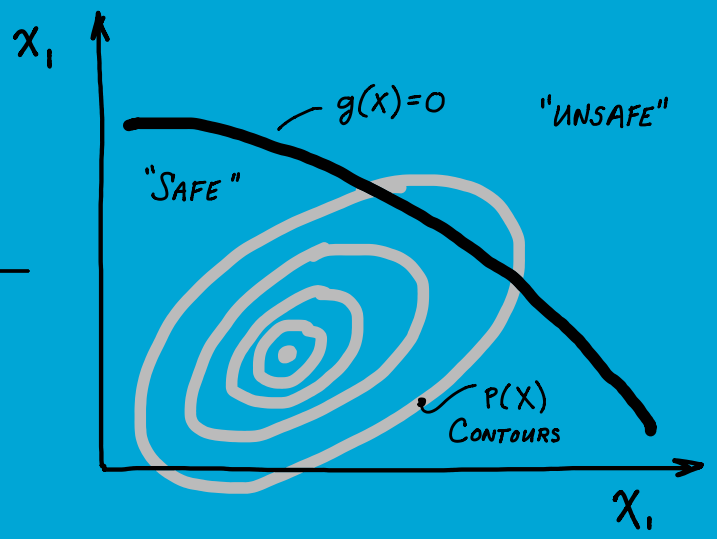
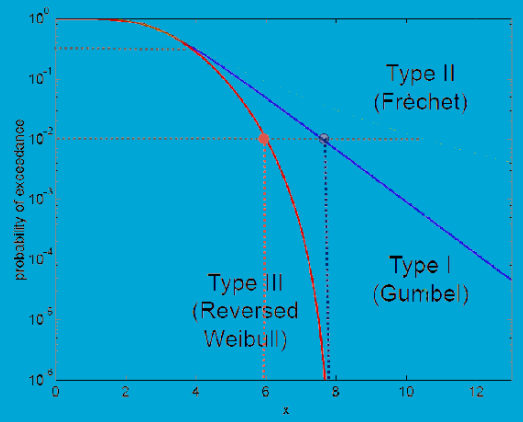
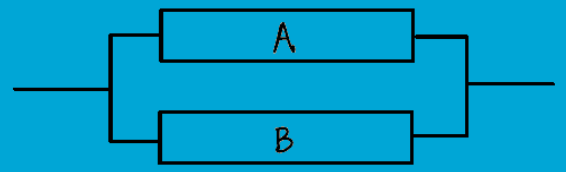
CIEM42X0 Probabilistic Design

Hydraulic and Offshore Structures (HOS) Track

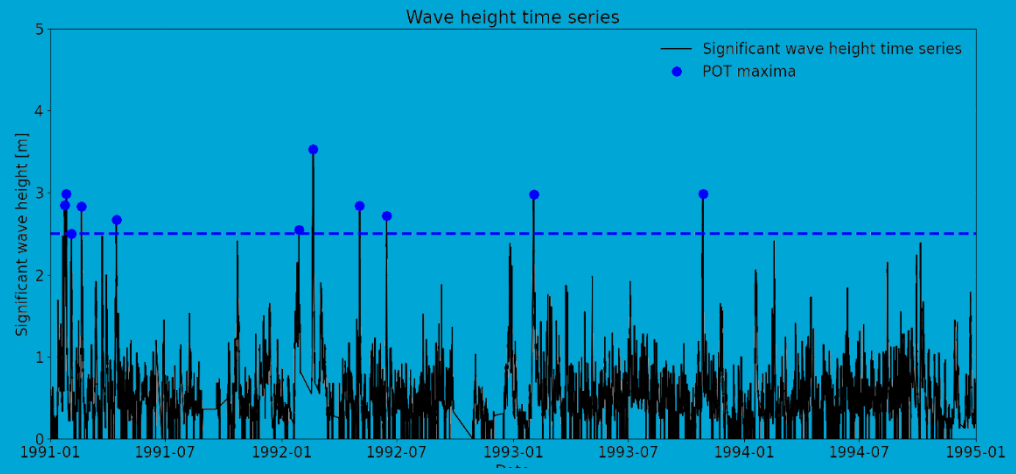
Civil Engineering MSc Program







Questions?



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